

CR-014: Admin Dashboard Statistics

1. Change Request Details

Field	Details
Change Request ID	CR-014
Title	Admin Dashboard Statistics
Priority	Medium
Difficulty	Medium
Change Type	Dashboard Enhancement
Affected Version	Student Management System v1.0 baseline
Dependency	Existing GET /api/students/ endpoint and authenticated Admin dashboard

2. Business Requirement

Administrators currently need to inspect the complete student list to understand the current size and recent registration activity of the Student Management System. The dashboard should provide a concise statistical overview so that administrators can understand key student information immediately after signing in.

The system should display reliable, automatically calculated student statistics without requiring administrators to manually count records or inspect registration dates.

3. Current Behaviour

- DashboardPage.jsx loads student records using the existing GET /api/students/ endpoint.
- The administrator dashboard primarily displays the student management table and existing controls.
- No dedicated summary statistics section is currently available.
- Administrators must inspect the student list manually to estimate the number of students and recent registrations.
- The existing student records include created_at information that can support date-based statistics.

4. Requested Change

Add an Admin Dashboard Statistics section above the administrator's student table.

The implementation must:

- Display the total number of students currently stored in the system.
- Display the number of students registered during the current calendar month.
- Display the number of students registered within the last seven days.
- Display the latest registered student's registration number and name when a student exists.
- Retrieve statistics through a dedicated authenticated dashboard API endpoint.
- Restrict the statistics endpoint to authenticated administrators.
- Display a suitable loading state while statistics are being retrieved.
- Handle empty student data without causing a dashboard error.
- Preserve all existing student CRUD, search, filtering, authentication, and authorization functionality.

5. Scope

In Scope

- Admin Dashboard statistics cards.
- Total student count.
- Current-month registration count.

- Last-seven-days registration count.
- Latest registered student information.
- Backend dashboard statistics API.
- Database aggregate and date-based queries using existing student records.
- Admin authentication and authorization for the new endpoint.
- Responsive presentation of the statistics section.
- Loading, empty-data, and API-error handling.

Out of Scope

- Changes to the existing student database structure.
- Changes to student registration or CRUD business rules.
- Changes to the existing client-side search and filtering functionality.
- Pagination or server-side sorting of the student list.
- New student roles or changes to the existing authentication model.
- Historical analytics, charts, or reporting beyond the four requested dashboard statistics.

6. Expected / Potentially Affected Files

- frontend/src/pages/DashboardPage.jsx
- frontend/src/styles/dashboard.css
- backend/src/app.js
- backend/src/controllers/dashboardController.js (new)
- backend/src/routes/dashboardRoutes.js (new)
- Existing authentication/authorization middleware, only where required to protect the new endpoint.

Students may modify additional files if required. Any additional files must be identified and justified in the Implementation Report.

7. Functional & Security Requirements

Functional Requirements

- FR-001: The administrator dashboard shall display a statistics section above the student list.
- FR-002: The system shall display the total number of students.
- FR-003: The system shall display the number of students registered during the current month.
- FR-004: The system shall display the number of students registered within the last seven days.
- FR-005: The system shall display the latest registered student's registration number and name when available.
- FR-006: Statistics shall be retrieved from the backend rather than hard-coded in the frontend.
- FR-007: The dashboard shall display a loading state while statistics are being retrieved.
- FR-008: The dashboard shall handle an empty student dataset without displaying invalid data.
- FR-009: Existing student CRUD and search/filter functionality shall continue to operate.

Non-Functional Requirements

- NFR-001: Statistics shall be calculated from current database records.
- NFR-002: The statistics section shall be responsive on desktop, tablet, and mobile screen sizes.
- NFR-003: The dashboard shall remain usable if the statistics request fails.
- NFR-004: The implementation shall avoid unnecessary API requests and database operations.

Security Requirements

- SEC-001: The dashboard statistics endpoint shall require valid authentication.
- SEC-002: Only users with the Admin role shall be authorized to access the statistics endpoint.
- SEC-003: The endpoint shall not accept client-supplied values to determine the reported student counts.
- SEC-004: Existing authentication and authorization mechanisms shall remain unchanged.

8. Acceptance Criteria

- AC-001: An authenticated administrator can view the statistics section on the dashboard.
- AC-002: Total Students matches the number of student records in the database.
- AC-003: This Month matches the number of students whose created_at date falls within the current calendar month.
- AC-004: Last 7 Days matches the number of students created within the defined seven-day period.
- AC-005: Latest Student displays the most recently registered student's registration number and name.
- AC-006: Statistics are retrieved successfully through the dashboard statistics API.
- AC-007: An unauthenticated request to the statistics endpoint is rejected.
- AC-008: A non-admin authenticated user is denied access to the statistics endpoint.
- AC-009: Empty student data is handled safely and does not break the dashboard.
- AC-010: Existing student CRUD, search/filter, and administrator authorization functionality continue to work after the change.
- AC-011: The statistics cards remain usable and readable on supported responsive screen sizes.

9. Expected Outcome

Administrators can immediately understand the current student population and recent registration activity from the dashboard without manually reviewing the complete student table. The new statistics are calculated from the existing student data and protected by the system's administrator authentication and authorization.

10. Implementation Report Requirement

After implementation, the Final Exam Software Change Implementation Report shall include:

- Implementation summary and technical approach.
- Actual affected files and any additional files introduced.
- Database query and API changes.
- Authentication and authorization verification.
- Functional, security, error-handling, and regression test cases.
- Acceptance criteria verification.
- Relevant screenshots and Postman/API evidence.
- Final status, limitations, and outstanding issues.

11. Change Control Notes

- The change is intended as an enhancement to the existing Admin Dashboard.
- No student records should be modified solely to implement this change.
- The existing GET /api/students/ endpoint and student CRUD operations shall remain available.
- The implementation should use existing student created_at data for date-based calculations.
- Any change to authentication or database schema beyond the approved scope requires separate review.